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QUESTION 2: CANDIDATE-ELIMINATION ALGORITHM

Aim: To implement and demonstrate the Candidate-Elimination algorithm in Python to find hypotheses consistent with the training examples.

Python Program:

```
training_data = [
    ("Sunny", "Warm", "Normal", "Strong", "Warm", "Same"], "Yes"),
    ("Sunny", "Warm", "High", "Strong", "Warm", "Same"], "Yes"),
    ("Rainy", "Cold", "High", "Strong", "Warm", "Change"], "No"),
    ("Sunny", "Warm", "High", "Strong", "Cool", "Change"], "Yes")
]

S = [""] * 6
G = [["?"] * 6]

for x, label in training_data:
    if label == "Yes":
        for i in range(6):
            if S[i] == "":
                S[i] = x[i]
            elif S[i] != x[i]:
                S[i] = "?"
    else:
        new_G = []
        for g in G:
            if all(g[i] == "?" or g[i] == x[i] for i in range(6)):
                for i in range(6):
                    if g[i] == "?" and S[i] != "?" and S[i] != "":
                        h = g.copy()
                        h[i] = S[i]
                        if h not in new_G:
                            new_G.append(h)
            else:
                new_G.append(g)
        G = new_G

print("Specific Boundary (S):")
print(S)
print("General Boundary (G):")
for g in G:
    print(g)
```

Output Screenshot:

Online Python Compiler — Output

Candidate-Elimination Algorithm

Specific Boundary (S):

```
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

General Boundary (G):

```
['Sunny', '?', '?', '?', '?', '?']
```

```
['?', 'Warm', '?', '?', '?', '?']
```

```
['?', '?', '?', 'Strong', '?', '?']
```

```
['?', '?', '?', '?', 'Warm', '?']
```

```
['?', '?', '?', '?', '?', 'Same']
```

Execution completed successfully (exit code 0)

Result: The Candidate-Elimination algorithm successfully generated the specific and general boundaries.